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What is Spring Scheduling ?
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=> Spring scheduling is a feature provided by the Spring Framework that allows you to schedule tasks or method to be executed at specific time intervals.

Ex :

```
** execute m1 ( ) method daily at 9:00 AM
** execute m2 ( ) method every monday @11:00 AM
** execute m3 ( ) method every Saturday @8:00 PM
```

=> It provides a convenient way to automate repetitive tasks, background processing, or any other operations that need to be executed periodically.

Spring scheduling is typically used in applications where you need to perform tasks at regular intervals, such as:

- 1) Generating reports or performing data backups at specific times.
- 2) Sending periodic notifications or reminders.
- 3) Updating cache or performing data sync.
- 4) Running Batch Jobs (hourly / daily / weekly / monthly / yearly)

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How to implement Scheduling in Spring Based Application ?
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=> By using `@EnableScheduling` and `@Scheduled` annotation we can implement scheduling in our spring application.

```
@Scheduled(fixedRate=1000)
public void m1(){

    // logic to generate report and send in email
}
```

fixedRate : This attribute specifies the interval between the start times of consecutive task executions. It means that the task will be executed at a fixed rate, regardless of the duration of the previous task execution.

fixedDelay : This attribute specifies the delay between the completion of the previous task execution and the start of the next execution. It means that the task will be executed with a fixed delay after the completion of the previous execution.

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What is cron expression ?
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=> Cron expression is used to define scheduling patterns for our tasks.

=> Cron expression provide a flexible and powerful way to specify the timing of task executions.

=> A cron expression consists of six fields, representing different aspects of the schedule. The fields are separated by spaces:

```
** Spring Cron Syntax : sec min hour day month weekday [year]
```

Position-1 : seconds (0 - 59)

Position-2 : Minutes (0 - 59)

Position-3 : Hours (0 - 23)

Position-4 : Day of month (1 - 31)

Position-5 : Month(1 - 12 or JAN-DEC)

Position-6 : Day of week (0 - 6 or SUN - SAT) (0 = sunday)

Position-7 : Year (empty, or 1970-2099) (optional)

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Every Second : * * * * *

Every Minute : 0 0/1 * * * * (or) 0 * * * * * (Ex: 9:10, 9:11, 9:12, 9:13...)

Every Hour : 0 0 * * * (Ex: 9:00 AM, 10:00 AM, 11:00 AM, 12:00 PM..)

Every Day at 9:00 AM : 0 0 9 * * *

Every day at 11:30 AM : 0 30 11 * * *

Every Monday 9:00 AM : 0 0 9 * * MON

Every Monday to Friday at 9:00 AM : 0 0 9 * * MON-FRI

Dec-31st at 12:00 AM : 0 0 0 31 12 *

02-Jun at 9:00 am every year : 0 0 9 2 JUN *

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